COLOSSEUM V2: Appendix

APPENDIX

The following contains additional details on the hardware and simulation results from the proposed benchmark. Note that throughout this report, **blue** indicates a visual variation, **green** indicates an action variation, and **red** indicates a language variation.

Note that in figs. 4 to 7 only tasks for which the `None` success rate is above 10% are included in the calculation for 'Mean % Change from None'. This reduces the noise of the statistic.

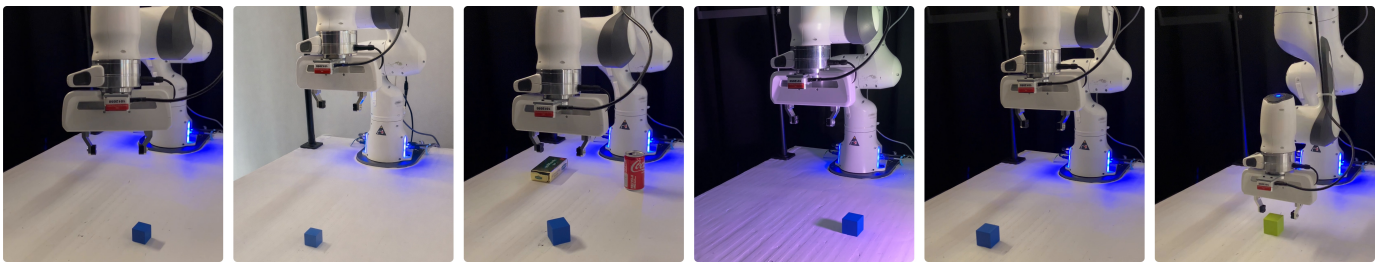


Fig. 1. Hardware setup with 5 variations for the RAISECUBE task. In order from left to right, the variations are `None`, `Background-Color`, `Distractor-Objects`, `Light-Color`, `MO-Size`, and `MO-Color`

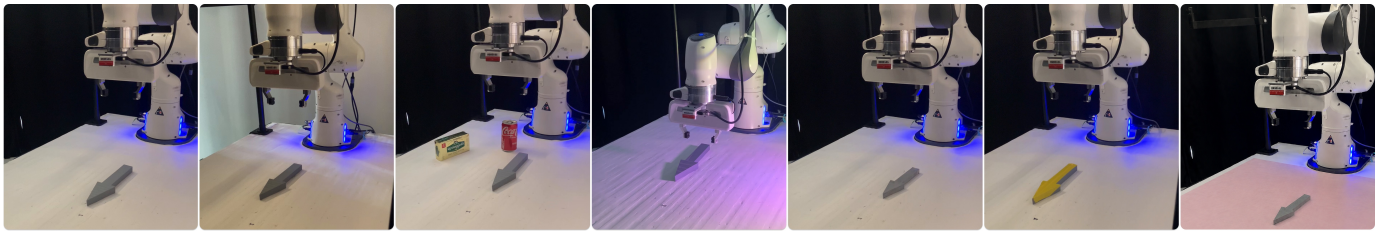


Fig. 2. Hardware setup with 5 variations for the ROTATEARROW task. In order from left to right, the variations are `None`, `Background-Color`, `Distractor-Objects`, `Light-Color`, `MO-Size`, and `MO-Color`

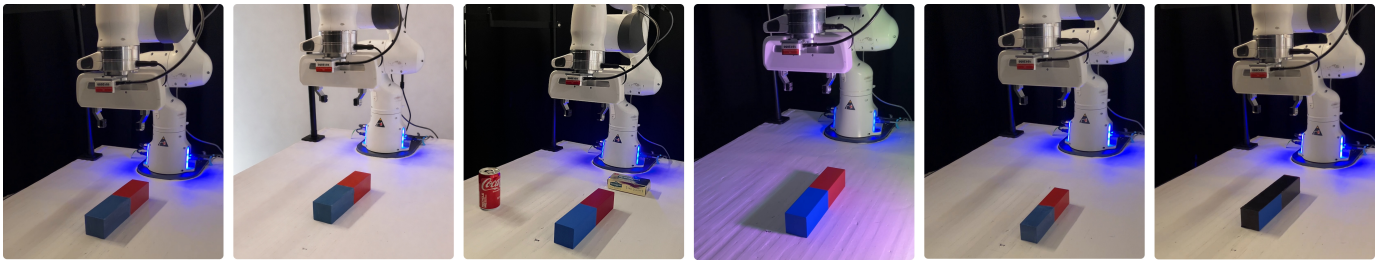


Fig. 3. Hardware setup with 5 variations for the LIFTPEGUPRIGHT task. In order from left to right, the variations are `None`, `Background-Color`, `Distractor-Objects`, `Light-Color`, `MO-Size`, and `MO-Color`

TASK	None	All	Mo color	Mo texture	Ro color	Ro texture	Table color	Light color	Table texture	Distractor object	Background texture	Background color	Camera pose	Pose randomization	Mo size	Ro size	Language paraphrase	Language other task	Language random	Language none
PickDishFromRack	100.0	0.5	94.0	58.0	52.5	66.0	0.0	45.0	18.0	87.5	98.5	100.0	94.0	-	-	-	100.0	100.0	100.0	100.0
RaiseCube	97.5	0.0	33.5	30.0	-	-	0.0	76.5	6.5	90.0	95.0	97.5	91.0	79.5	95.5	-	98.0	98.5	98.5	97.0
RotateArrow	96.5	0.0	42.0	22.0	-	-	0.0	10.0	1.0	53.5	63.0	90.5	94.5	71.5	84.5	-	95.5	97.0	96.5	97.0
LiftPegUpright	56.5	0.0	28.5	5.0	-	-	0.0	26.0	4.5	31.0	29.5	32.0	48.5	41.0	20.5	-	51.5	61.5	52.5	57.0
HammerNail	46.0	3.0	7.0	1.0	62.0	28.0	0.0	0.0	0.0	27.0	18.0	18.0	9.0	31.0	32.0	-	49.0	47.0	47.0	49.0
PickSodaFromCabinet	31.5	0.0	1.0	2.5	7.5	1.0	11.0	3.5	24.0	22.5	17.5	38.5	9.0	18.0	-	-	32.5	13.5	22.5	26.5
ScoopBanana	18.0	6.0	10.0	2.0	22.0	15.0	1.0	20.0	6.0	16.0	10.0	10.0	18.0	20.0	22.0	10.0	20.0	14.0	20.0	16.0
PlaceDishInRack	11.0	0.0	8.5	0.5	11.0	13.0	0.0	6.0	1.0	8.5	17.5	20.5	6.0	8.0	-	-	10.0	7.0	5.0	8.0
StackCube	7.5	0.0	5.0	1.5	-	-	0.0	2.0	0.0	5.5	14.0	10.5	14.0	2.0	10.5	9.5	10.0	5.5	10.0	12.0
PegInsertionSide	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.0	0.0	-	-	0.0	0.0	0.0	0.0
OpenDrawer	1.0	-	-	-	-	-	0.0	0.0	1.0	7.0	0.0	0.0	0.0	1.0	-	-	1.0	2.0	0.0	3.0
PlaceBookInShelf	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0
PlugCharger	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0
OpenCabinet	0.0	3.0	-	-	-	-	0.0	0.0	0.0	10.0	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0
PlaceCubeInDrawer	0.0	0.0	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	0.0	0.0	0.0	0.0
CookItemInPan	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Mean % Change from None	0.0	-95.0	-53.3	-81.7	-13.3	-33.7	-94.9	-55.4	-81.0	-26.7	-22.2	-5.8	-28.3	-23.3	-17.3	-44.4	0.3	-12.9	-9.5	-5.9
Mean Success Rate	29.2	0.8	17.7	9.4	17.2	13.7	0.8	11.8	3.9	22.4	22.7	26.1	24.1	18.1	33.1	6.5	29.2	27.9	28.2	29.1

Fig. 4. Raw results for ACT on the Single-Arm task suite

TASK	None	All	Mo color	Mo texture	Ro color	Ro texture	Table color	Light color	Table texture	Distractor object	Background texture	Background color	Camera pose	Pose randomization	Mo size	Ro size	Language paraphrase	Language other task	Language random	Language none
DualArmLiftPot	97.0	1.0	68.0	23.0	-	-	78.0	24.0	24.0	78.0	89.0	80.0	78.0	87.0	47.0	-	89.0	94.0	92.0	97.0
DualArmPushBox	87.0	1.0	74.0	71.0	76.0	64.0	67.0	75.0	24.0	67.0	82.0	89.0	52.0	65.0	67.0	83.0	82.0	85.0	89.0	87.0
DualArmCubeHandover	84.0	0.5	42.5	32.0	-	-	19.5	31.0	4.5	3.0	0.0	0.0	1.0	46.5	82.5	-	88.0	88.5	90.5	84.0
DualArmLiftTray	81.0	0.0	92.0	62.0	-	-	61.0	75.0	35.0	70.0	78.0	83.0	50.0	61.0	15.0	-	79.0	78.0	76.0	80.0
DualArmThreading	72.0	0.0	45.0	39.0	57.0	48.0	9.0	36.0	2.0	15.0	51.0	47.0	48.0	30.0	-	-	67.0	63.0	69.0	60.0
DualArmPourPot	61.0	3.0	62.0	24.0	62.0	54.0	48.0	34.0	61.0	50.0	64.0	55.0	55.0	24.0	24.0	59.0	53.0	58.0	60.0	56.0
DualArmStackCube	49.0	0.0	6.0	4.0	41.0	47.0	7.0	18.0	4.0	18.0	31.0	40.0	4.0	22.0	35.0	57.0	45.0	37.0	50.0	50.0
DualArmPenCap	2.0	0.0	2.0	5.0	1.0	6.0	4.0	2.0	1.0	3.0	4.0	1.0	3.0	1.0	-	-	3.0	1.0	10.0	2.0
DualArmStackTwoCubes	1.0	0.0	2.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	1.0	1.0	0.0	0.0	6.0	1.0	8.0	3.0	2.0
DualArmBottleHandover	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0	1.0	1.0	0.0	0.0	0.0	-	0.0	0.0	0.0	0.0
DualArmDrawerPlace	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	0.0	0.0	0.0	0.0
DualArmDrawerOpen	0.0	0.0	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0
Mean % Change from None	0.0	-98.9	-29.2	-54.1	-12.0	-18.8	-48.4	-45.3	-69.7	-44.7	-25.5	-25.1	-47.4	-39.9	-41.2	2.8	-5.7	-6.5	-0.7	-3.4
Mean Success Rate	44.5	0.5	35.8	23.6	39.5	36.5	24.5	24.6	13.0	25.3	33.4	33.1	24.3	28.0	30.1	51.2	42.2	42.7	45.0	43.2

Fig. 5. Raw results for ACT on the Bimanual task suite

TASK	None	All	Mo color	Mo texture	Ro color	Ro texture	Table color	Light color	Table texture	Distractor object	Background texture	Background color	Camera pose	Pose randomization	Mo size	Ro size	Language paraphrase	Language other task	Language random	Language none
PickDishFromRack	100.0	21.5	71.0	79.0	93.0	85.0	88.0	95.0	59.0	89.0	95.0	87.0	82.0	-	-	-	80.0	34.0	37.0	86.0
RotateArrow	86.0	2.0	59.0	62.0	-	-	68.0	76.0	44.0	65.0	78.0	85.0	73.0	62.0	80.0	-	40.5	0.0	0.0	0.0
RaiseCube	73.0	12.5	63.5	65.5	-	-	48.0	68.5	53.0	72.5	81.5	77.5	68.5	55.0	71.0	-	67.5	25.0	38.0	66.0
OpenDrawer	25.0	13.5	-	-	-	-	22.0	21.0	29.0	20.0	29.0	29.0	32.0	25.0	-	-	37.5	8.5	4.0	11.0
ScoopBanana	21.0	12.5	18.0	17.0	19.0	24.0	16.0	21.0	19.0	27.0	21.0	20.0	21.0	22.0	22.0	21.0	10.0	3.0	3.0	2.0
PickSodaFromCabinet	11.0	2.5	9.0	14.0	6.0	4.0	12.0	5.0	8.0	14.0	9.0	2.0	10.0	-	-	-	22.5	0.0	0.0	0.0
HammerNail	7.0	4.0	14.0	16.0	14.0	14.0	16.0	16.0	6.0	14.0	13.0	12.0	5.0	8.0	6.0	-	14.5	2.0	0.0	1.0
StackCube	6.0	0.0	1.0	1.0	-	-	3.0	3.0	1.0	6.0	10.0	10.0	8.0	6.0	7.0	6.0	5.5	0.0	0.0	2.5
LiftPegUpright	4.0	0.0	2.0	0.0	-	-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	2.0	1.0	-	7.0	1.5	0.0	0.0
PlaceDishInRack	3.0	0.5	7.0	4.0	7.0	7.0	8.0	8.0	5.0	7.0	8.0	10.0	8.0	6.0	-	-	3.0	0.0	0.0	2.5
PlaceBookInShelf	2.0	0.5	1.0	3.0	3.0	1.0	3.0	5.0	4.0	3.0	3.0	2.0	3.0	1.0	-	-	6.0	1.5	0.0	5.0
PlaceCubeInDrawer	1.0	0.0	-	-	-	-	0.0	1.0	1.0	1.0	0.0	0.0	0.0	1.0	1.0	-	0.0	0.0	0.0	0.0
PegInsertionSide	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0
PlugCharger	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0
OpenCabinet	0.0	0.5	-	-	-	-	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0
CookItemInPan	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0
Mean % Change from None	0.0	-70.5	-21.2	-10.2	-20.7	-21.5	-15.6	-15.6	-23.0	-0.0	-0.8	-13.1	-3.4	-12.0	-1.7	0.0	3.6	-80.6	-80.1	-61.7
Mean Success Rate	21.2	4.4	18.9	20.1	15.8	15.0	17.8	20.0	14.3	20.0	21.7	20.9	19.4	13.4	23.6	9.0	18.4	4.7	5.1	11.0

Fig. 6. Raw results for $\pi_{0.5}$on the Single-Arm task suite

TASK	None	All	Mo color	Mo texture	Ro color	Ro texture	Table color	Light color	Table texture	Distractor object	Background texture	Background color	Camera pose	Pose randomization	Mo size	Ro size	Language paraphrase	Language other task	Language random	Language none
DualArmPushBox	56.0	0.0	51.0	40.0	38.0	33.0	37.0	55.0	13.0	46.0	57.0	60.0	46.0	29.0	58.0	56.0	10.0	0.0	0.0	0.0
DualArmPourPot	41.0	3.5	23.0	26.0	51.0	38.0	21.0	34.0	52.0	23.0	33.0	35.0	42.0	12.0	17.0	41.0	30.5	20.0	3.5	21.0
DualArmCubeHandover	27.0	0.0	10.0	11.0	-	-	12.0	31.0	5.0	11.0	31.0	34.0	21.0	16.0	17.0	-	0.0	0.0	0.0	0.0
DualArmLiftTray	15.0	0.0	4.0	3.0	-	-	19.0	7.0	2.0	4.0	0.0	1.0	15.0	9.0	5.0	-	8.0	0.0	4.0	9.0
DualArmLiftPot	5.0	5.0	9.0	6.0	-	-	7.0	4.0	6.0	6.0	14.0	18.0	5.0	6.0	6.0	-	4.0	4.0	12.5	9.5
DualArmPenCap	4.0	5.5	1.0	1.0	3.0	3.0	1.0	2.0	1.0	1.0	2.0	2.0	2.0	6.0	-	-	0.0	0.0	0.0	0.0
DualArmStackCube	3.0	0.0	0.0	0.0	1.0	2.0	1.0	4.0	0.0	0.0	0.0	0.0	9.0	1.0	0.0	3.0	0.0	0.0	0.0	3.0
DualArmBottleHandover	0.0	0.0	3.0	3.0	-	-	1.0	3.0	1.0	1.0	1.0	1.0	2.0	2.0	3.0	-	10.5	4.5	0.0	4.0
DualArmThreading	0.0	0.0	0.0	0.0	1.0	1.0	4.0	1.0	0.0	2.0	0.0	2.0	1.0	0.0	-	-	0.0	0.0	3.0	0.0
DualArmDrawerPlace	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	0.0	0.0	0.0	0.0
DualArmDrawerOpen	0.0	0.0	-	-	-	-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0
DualArmStackTwoCubes	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Mean % Change from None	0.0	-97.9	-47.3	-51.1	-3.9	-24.2	-27.9	-14.3	-54.5	-48.6	-25.7	-18.7	-9.4	-49.9	-39.7	0.0	-63.6	-87.8	-91.2	-72.2
Mean Success Rate	12.6	1.2	9.2	8.2	15.7	12.8	8.6	11.8	6.7	7.8	11.6	12.8	11.9	6.8	11.8	25.0	5.2	2.4	1.9	3.9

Fig. 7. Raw results for $\pi_{0.5}$on the Bimanual task suite

TASK	None	All	Mo color	Mo texture	Ro color	Ro texture	Table color	Light color	Table texture	Distractor object	Background texture	Background color	Camera pose	Pose randomization	Mo size	Ro size	Language paraphrase	Language other task	Language random	Language none
RotateArrow	96.0	6.0	59.0	38.0	-	-	39.0	86.0	29.0	46.0	72.0	86.5	91.0	62.0	79.0	-	90.0	79.0	83.5	90.5
RaiseCube	90.5	19.5	79.5	74.0	-	-	64.5	87.0	48.5	77.5	88.5	91.0	61.5	83.5	89.0	-	87.0	73.5	76.0	72.5
PickDishFromRack	60.5	18.0	97.0	96.0	78.5	70.5	57.5	71.0	48.5	50.5	82.0	89.0	65.0	-	-	-	61.0	71.0	92.0	97.0
HammerNail	28.5	1.5	5.5	5.0	21.5	15.0	0.0	20.0	4.5	20.0	30.0	32.5	9.0	25.0	30.0	-	24.0	20.0	18.5	14.0
ScoopBanana	16.0	6.0	4.0	8.0	8.0	8.0	-	-	-	10.5	-	-	-	-	16.0	24.0	-	-	-	-
PlaceBookInShelf	11.0	0.0	1.5	5.0	0.0	6.0	8.5	3.0	10.0	12.0	10.5	7.5	0.5	4.0	-	-	13.0	7.0	6.0	2.0
PlaceDishInRack	6.0	0.0	17.0	6.0	9.0	10.0	3.5	7.5	8.5	4.0	4.0	6.0	7.0	3.0	-	-	6.0	7.0	7.0	3.5
PickSodaFromCabinet	3.0	0.0	0.0	0.5	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.5	0.5	-	-	-	1.0	0.0	0.0	0.0
LiftPegUpright	2.0	0.0	2.0	0.0	-	-	4.0	8.0	0.0	0.0	2.0	0.0	3.0	8.0	1.0	-	4.0	3.0	6.0	5.0
StackCube	1.5	0.0	0.5	1.0	-	-	0.0	0.5	1.0	0.5	2.0	3.0	1.5	0.5	1.0	1.5	2.5	3.0	3.0	2.5
PlugCharger	1.5	0.0	0.0	0.0	0.0	0.0	0.0	1.5	0.0	0.0	0.0	0.0	0.0	0.0	-	-	1.5	0.0	2.0	0.0
PegInsertionSide	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0
Mean % Change from None	0.0	-83.3	-38.7	-34.5	-36.2	-31.6	-43.2	-19.9	-45.9	-23.0	1.8	4.0	-38.7	-29.8	-3.5	50.0	-1.4	-17.1	-11.5	-19.6

Fig. 8. Raw results for MolmoAct2on the Single-Arm task suite

TASK	None	All	Mo color	Mo texture	Ro color	Ro texture	Table color	Light color	Table texture	Distractor object	Background texture	Background color	Camera pose	Pose randomization	Mo size	Ro size	Language paraphrase	Language other task	Language random	Language none
DualArmPourPot	22.5	1.0	2.0	0.0	3.5	10.0	10.0	0.0	2.5	13.5	12.5	16.5	3.5	13.5	7.5	20.0	5.5	23.5	9.0	26.5
DualArmLiftPot	13.5	1.0	4.5	5.0	-	-	16.0	10.0	3.0	16.5	17.5	18.5	16.5	11.5	14.0	-	15.0	28.0	20.0	14.0
DualArmPushBox	6.5	0.0	0.0	0.0	5.0	8.0	2.0	6.5	3.0	5.0	3.5	4.5	16.5	2.5	6.5	3.5	5.0	3.5	1.5	3.5
DualArmCubeHandover	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	0.0	0.0	0.0	0.0
DualArmBottleHandover	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	0.0	0.0	0.0	0.0
DualArmLiftTray	0.0	0.0	0.0	0.0	-	-	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	0.0	0.0	0.0	0.0
DualArmThreading	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	-	-	-	0.0	0.0	-	-
DualArmPenCap	-	3.0	-	-	-	-	0.0	-	0.0	-	-	-	-	-	-	-	-	-	-	-
DualArmDrawerPlace	-	0.0	-	-	-	-	0.0	-	0.0	-	-	-	-	-	-	-	-	-	-	-
DualArmDrawerOpen	-	0.0	-	-	-	-	0.0	-	0.0	-	-	-	-	-	-	-	-	-	-	-
DualArmStackCube	-	0.0	-	-	-	-	0.0	-	0.0	-	-	-	-	-	-	-	-	-	-	-
Mean % Change from None	0.0	-94.1	-78.9	-81.5	-84.4	-55.6	-18.5	-63.0	-83.3	-8.9	-7.4	5.2	-31.1	-27.4	-31.5	-11.1	-32.2	55.9	-5.9	10.7

Fig. 9. Raw results for MolmoAct2on the Bimanual task suite

Model	Included tasks	Excluded tasks
ACT - Single-Arm	HAMMERNAIL, LIFTPEGUPRIGHT, PICKDISHFROMRACK, PICKSODAFROMCABINET, PLACEDISHINRACK, RAISECUBE, ROTATEARROW, SCOOPBANANA	COOKITEMINPAN, OPENCABINET, OPENDRAWER, PEGINSERTIONSIDE, PLACEBOOKINSHELF, PLACECUBEINDRAWER, PLUGCHARGER, STACKCUBE
ACT - Bimanual	DUALARMCUBEHANDOVER, DUALARMLIFTPOT, DUALARMLIFTTRAY, DUALARMPOURPOT, DUALARMPUSHBOX, DUALARMSTACKCUBE, DUALARMTHREADING	DUALARMBOTTLEHANDOVER, DUALARMDRAWEROPEN, DUALARMDRAWERPLACE, DUALARMPENCAP, DUALARMSTACKTWOCUBES
Pi0.5 - Single-Arm	OPENDRAWER, PICKDISHFROMRACK, PICKSODAFROMCABINET, RAISECUBE, ROTATEARROW, SCOOPBANANA	COOKITEMINPAN, HAMMERNAIL, LIFTPEGUPRIGHT, OPENCABINET, PEGINSERTIONSIDE, PLACEBOOKINSHELF, PLACECUBEINDRAWER, PLACEDISHINRACK, PLUGCHARGER, STACKCUBE
Pi0.5 - Bimanual	DUALARMCUBEHANDOVER, DUALARMLIFTTRAY, DUALARMPOURPOT, DUALARMPUSHBOX	DUALARMBOTTLEHANDOVER, DUALARMDRAWEROPEN, DUALARMDRAWERPLACE, DUALARMLIFTPOT, DUALARMPENCAP, DUALARMSTACKCUBE, DUALARMSTACKTWOCUBES, DUALARMTHREADING
MolmoAct2 - Single-Arm	HAMMERNAIL, PICKDISHFROMRACK, PLACEBOOKINSHELF, RAISECUBE, ROTATEARROW, SCOOPBANANA	LIFTPEGUPRIGHT, PEGINSERTIONSIDE, PICKSODAFROMCABINET, PLACEDISHINRACK, PLUGCHARGER, STACKCUBE
MolmoAct2 - Bimanual	DUALARMLIFTPOT, DUALARMPOURPOT	DUALARMBOTTLEHANDOVER, DUALARMCUBEHANDOVER, DUALARMLIFTTRAY, DUALARMPUSHBOX, DUALARMTHREADING

TABLE I
TASKS INCLUDED AND EXCLUDED WHEN CALCULATING THE AVERAGE CHANGE IN SUCCESS RATE FOR EACH PERTURBATION.